



Human Rhinovirus Leverages Multi-Layered Immune Evasion Mechanisms To Bypass Host Defenses

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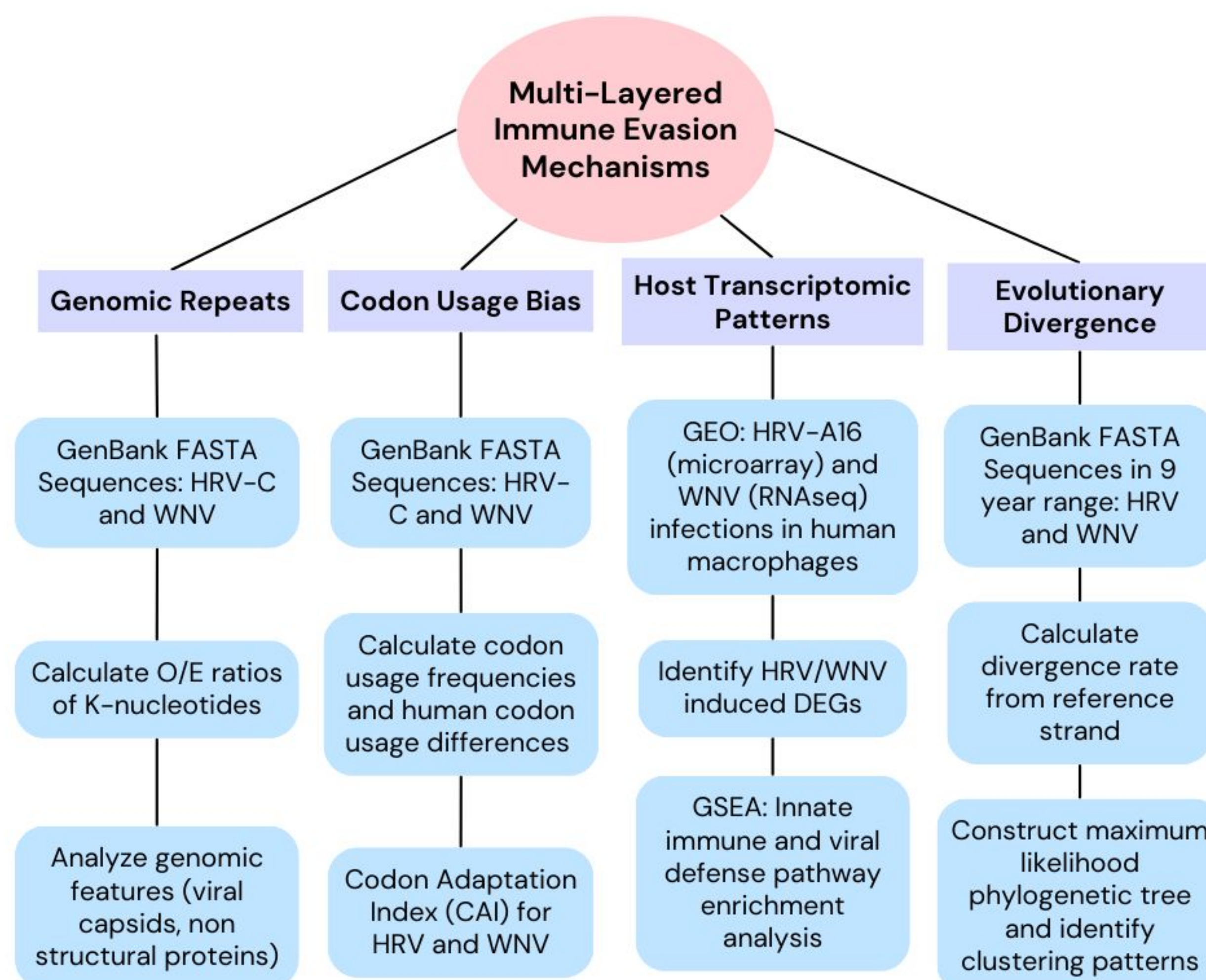
ABSTRACT

- Human rhinovirus (HRV), the leading cause of the common cold, remains a persistent global pathogen despite widespread exposure and repeated immune priming.
- With over 160 recognized types, HRV evades lasting immune memory and continues to cause recurrent infections throughout life. Despite extensive research, no vaccine has been developed due to its extensive and multi-faceted immune evasion mechanisms.
- In this study, we examine HRV's immune evasion strategies through four key layers: (1) genome composition, focusing on CpG suppression; (2) codon usage bias and its implications for immune recognition; (3) host transcriptional responses to HRV infection; and (4) evolutionary dynamics and sequence variability.
- Our results reveal a set of interlocking immune evasion strategies that reinforce each other, including region-specific CpG suppression and TpG enrichment, codon bias driven by immune avoidance rather than translational efficiency, muted innate immune activation, and heightened sequence divergence likely shaped by human selective pressures.
- Our findings support a paradigm shift toward mechanism-based frameworks in virology, guiding vaccine design and antiviral strategies beyond pathogen-specific models.

BACKGROUND

- Human rhinovirus (HRV) is a positive-sense single-stranded RNA virus that is the most common cause of upper respiratory tract infections like the common cold.
- Development of effective vaccines has been hindered by HRV's extensive genomic diversity and capacity to evade host immune responses.
- This phenomenon suggests that HRV employs multiple, effective strategies to evade human immune responses, but the extent to which these interact is not well characterized.
- Understanding the molecular and evolutionary mechanisms HRV uses to escape immune detection may provide insight into persistent infection and immune evasion strategies shared across RNA viruses and guide vaccine efforts..
- To contextualize our findings, we compare HRV with West Nile Virus (WNV), a structurally similar flavivirus that is a dead-end host in humans and displays distinct patterns of immune engagement.

METHODS



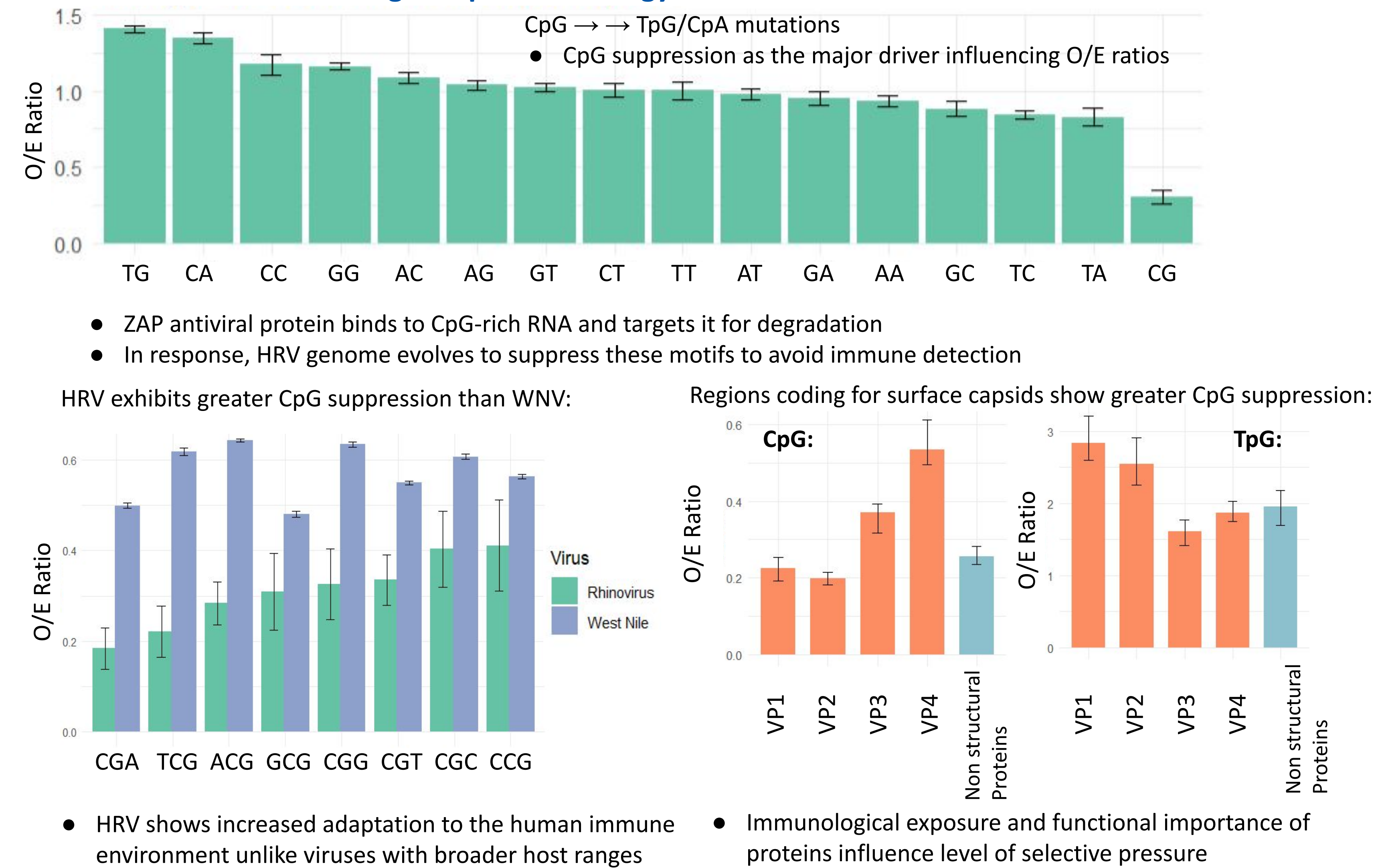
• Identify individual molecular signatures of immune avoidance in HRV

• Reveal how different evasion tactics may reinforce one another

• Evaluate whether HRV's success is driven by a single dominant strategy or a coordinated set of immune evasion mechanisms

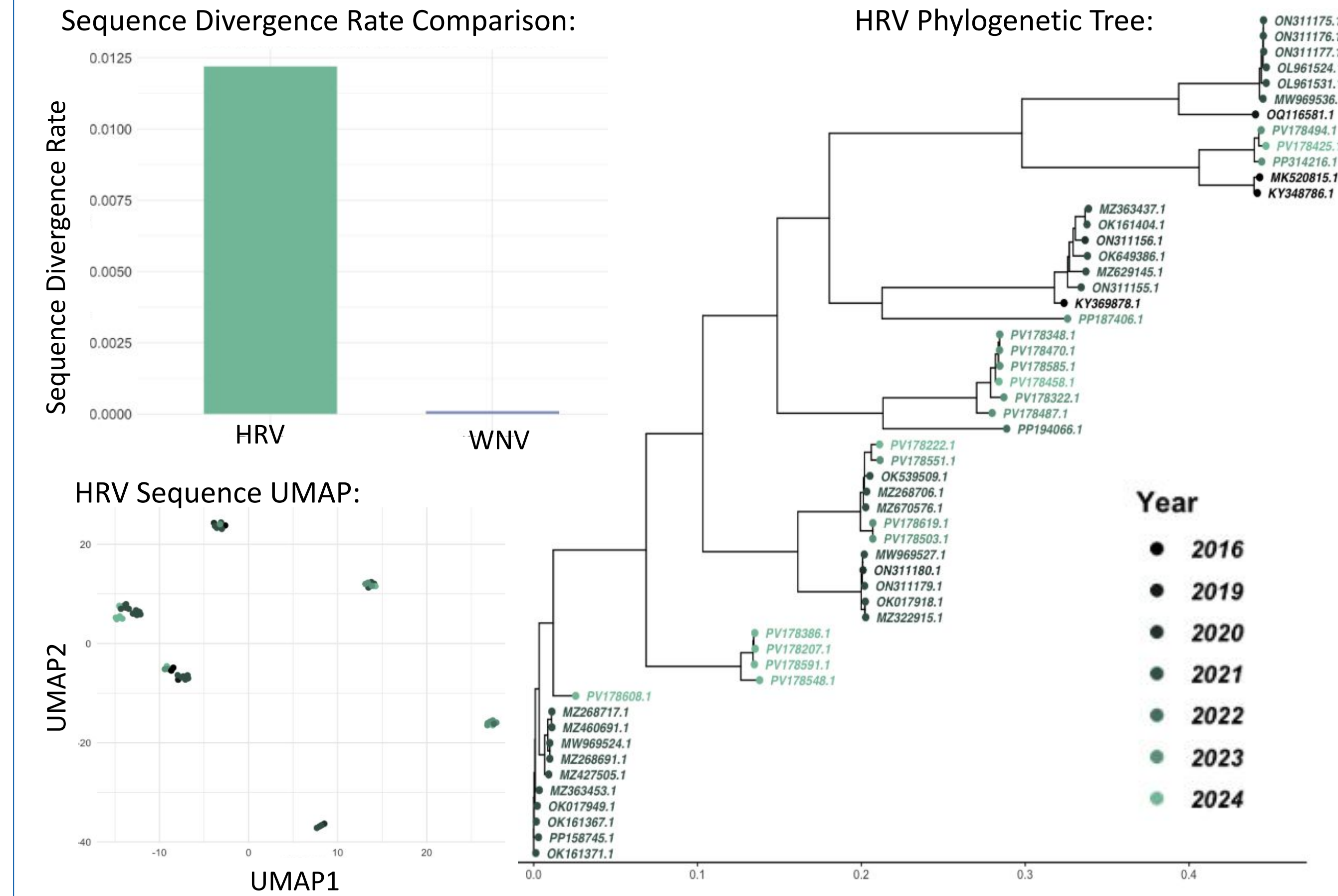
RESULTS

CpG Suppression and TpG Enrichment in Rhinovirus Genomes Function as a Coordinated, Region-specific Strategy for Immune Evasion



RESULTS (cont)

Persistent Human Selective Pressures May Underlie Greater Sequence Divergence in Rhinovirus



- HRV exhibits higher sequence divergence rate than WNV, highlighting its rapid evolution as a potential mechanism for evading host treatments.
- The observed year-by-year clustering in both the phylogenetic tree and UMAP analysis indicates that HRV evolves progressively over time, rather than mutations arising randomly.
- Persistent exposure to host immune responses and antiviral treatments may impose selective pressures that contribute to the higher sequence divergence rate.

CONCLUSION

- Our findings reveal that HRV employs a coordinated set of molecular tactics to evade innate immune detection and sustain replication in human hosts.
- Together, these findings form a complementary and interconnected framework of immune evasion. Rather than acting independently, these mechanisms reinforce each other: CpG suppression reduces host sensor recognition and shapes codon usage, while mutation-driven variation enhances the virus's ability to escape immune detection.
- Our analysis was constrained by several limitations:
 - For genomic/codon analyses, we were limited to 16 HRV strains and 12 WNV strains, which may not capture the full genetic diversity.
 - CAI values were calculated using average human codon frequencies, which may not reflect the specific codon preferences of HRV's main target: respiratory epithelial cells
 - Host responses were measured using different platforms (microarray v. RNAseq)
 - Sequence divergence was calculated from different time spans & unequal strain representation
 - While WNV served as a useful contrast due to its phylogenetic distance and divergent host ecology, its rarity and zoonotic lifestyle could introduce confounding variables
- Our study underscores the need to shift from pathogen-specific to mechanism-based frameworks in antiviral research, guiding interventions that can outmaneuver even the most evasive viruses.

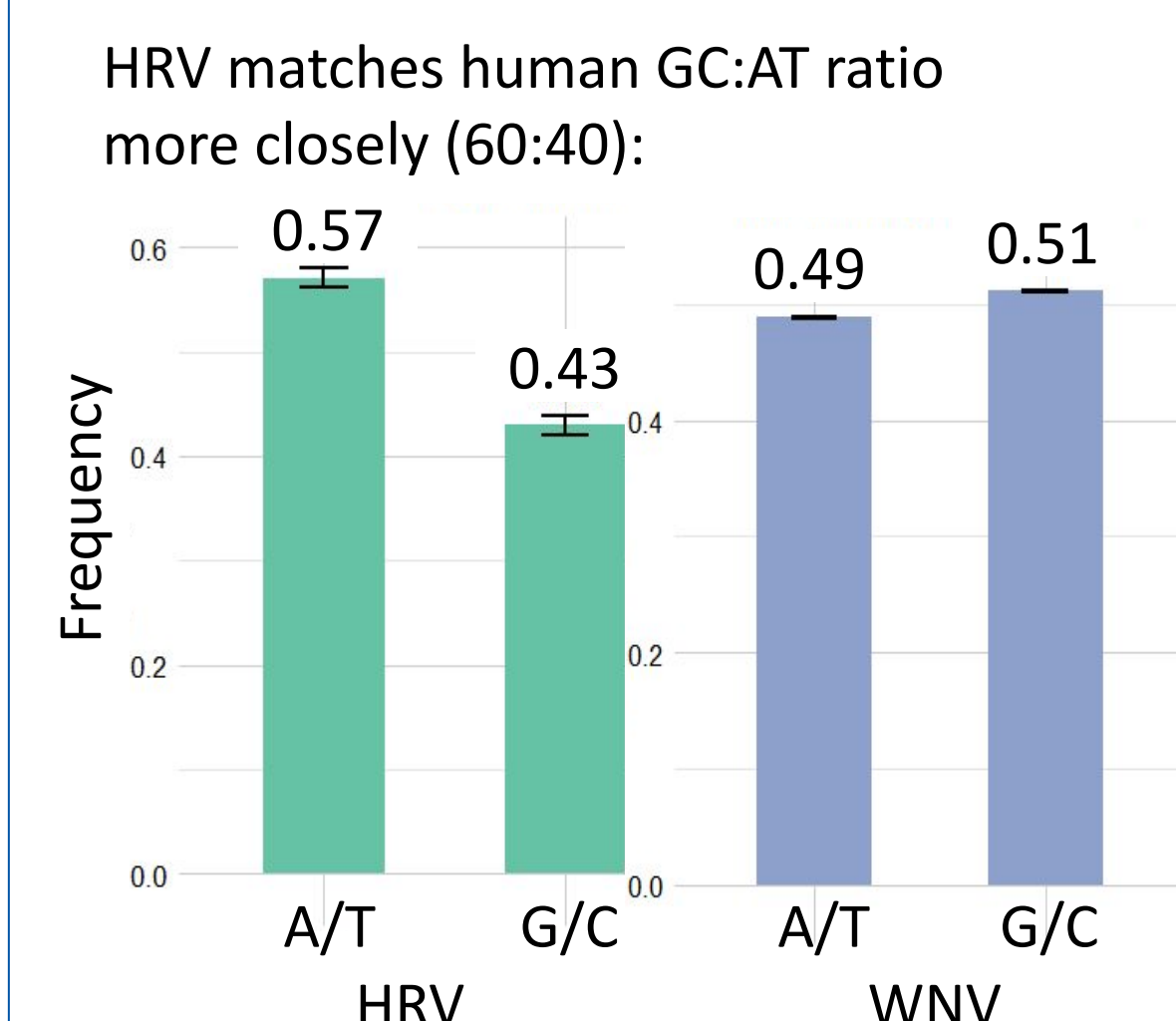
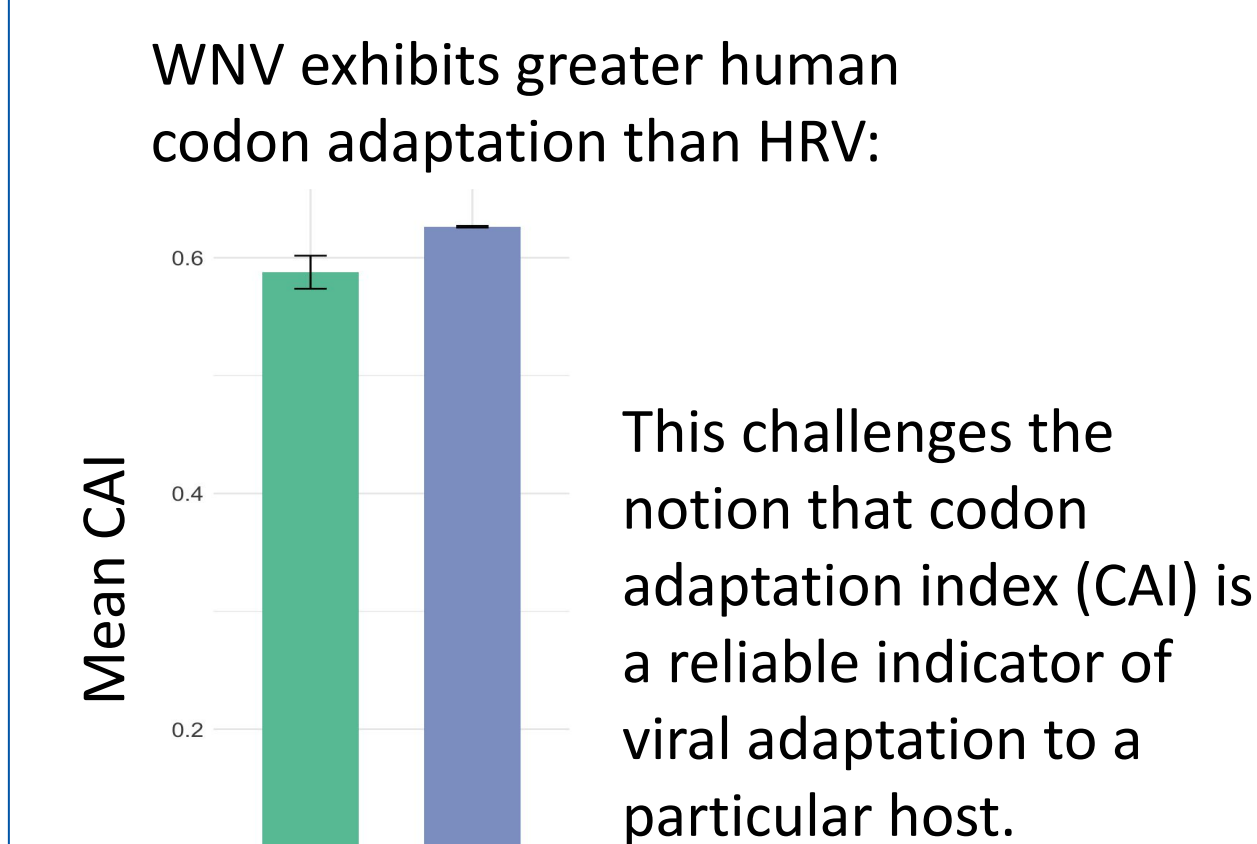
FUTURE STEPS

- Conduct a quantitative study across multiple viruses (focusing on human respiratory viruses, e.g. seasonal influenza) to validate CpG suppression as a broad marker of immune evasion strength.
- Adopt unsupervised or machine learning approaches for pathway enrichment and host response profiling to reveal previously unrecognized immune signatures.
- Expand temporal and strain-based sequence divergence analysis across several viruses
- Integrate transcriptional response with GWAS to identify host genetic variants influencing susceptibility to HRV infection.
- Integrate various metrics (CpG suppression levels, CAI deviations, transcriptional immune response profiles, divergence rates) to generate a scalable model to measure immune evasion strength.

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Immune Avoidance, Not Translation Efficiency, Drives Codon Bias



- Viruses with host-like nucleotide mimicry can evade immune detection of foreign RNA.
- CAI doesn't account for evolutionary pressures like CpG suppression, nucleotide composition, and host immune recognition.

Subdued Innate Immune Response to HRV Reflects Host-Adapted Immune Evasion Strategy

